

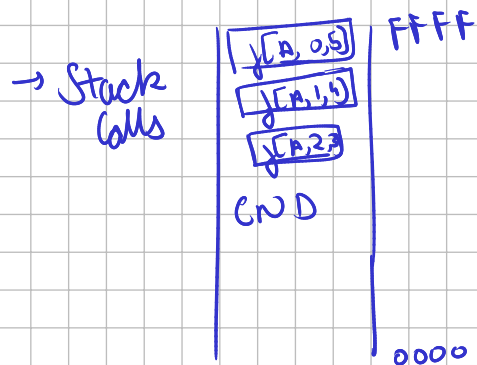
RECURSION PROBLEMS:

1) INVERSE ARRAY:



$l \rightarrow$ left pointer
 $r \rightarrow$ right pointer

$\Rightarrow$ So $f[A, l, r]$:
 $\circ$ if $(l < r)$:
 Swap $[A[l], A[r]]$;
 $f[A, l+1, r-1]$



2) Check for palindrome:

So, eg $A C Q C A$ so, use two pointer;
 $\uparrow \quad \uparrow$
 = same

$f(S, l, r)$:
 if $(l < r)$:
 if $(S[l] \neq S[r])$:
 return false;
 else:
 return $f(S, l+1, r-1)$;
 else:
 return true;

eg
 0 1 2 3 4
 M A D A M
 $\uparrow \quad \uparrow$
 $l \quad r$

$\Rightarrow 0 < 4$
 $M = M$ (1st call)
 $1 < 3$ (2nd call)
 $A = A$
 $l = r$ (3rd call)
 return true;

$\therefore$ notice that in recursion, we rely on a breaking change, i.e., something that can determine the final answer immediately.
 Eg, in given palindrome problem, it was unequal char at i^{th} & $n-i^{th}$ position

MULTIPLE RECURSION CAUSE:

Eg) Consider Fibonacci Series;

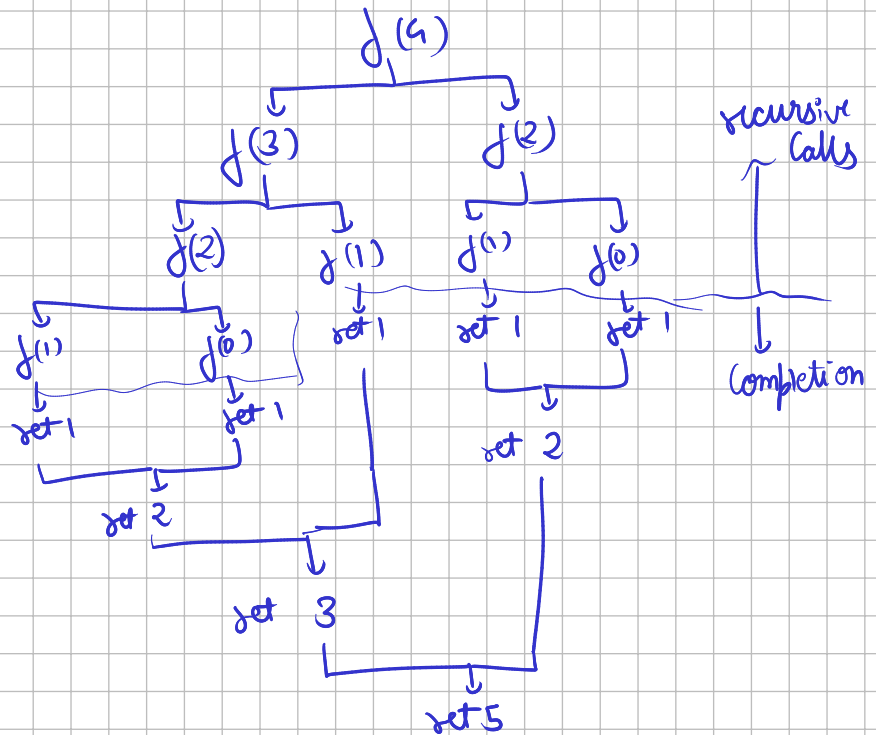
$$f(0)=1, f(1)=1, f(2)=2, f(3)=3, f(4)=5, f(5)=8 \dots$$

$$So, f(n) = f(n-1) + f(n-2) \quad \forall n > 2$$

$f[n]$:

$\text{if } n \leq 1;$
 $\text{return } 1;$

$A = f[n-1]$
 $B = f[n-2]$
 $\text{return } A+B;$



$$\therefore f(4) = 5$$

So it is kind of like:

$f(n)$
 $\{$
 $\quad A = f(n-1) \{$
 $\quad \quad A = f(n-2) \{$
 $\quad \quad \quad \dots$
 $\quad \quad \quad \{ \text{ret}$
 $\quad \quad B = f(n-2) \{$
 $\quad \quad \quad \dots$
 $\quad \quad \quad \{ \text{ret}$
 $\quad \quad \quad \{ \text{ret}$
 $\quad B = f(n-2) \{$

$$A = f(n-3) \{$$

...
 $\}$ set

$$B = f(n-4) \{$$

...
 $\}$ set

$\}$ set

$$\text{set } A+B;$$

$\}$